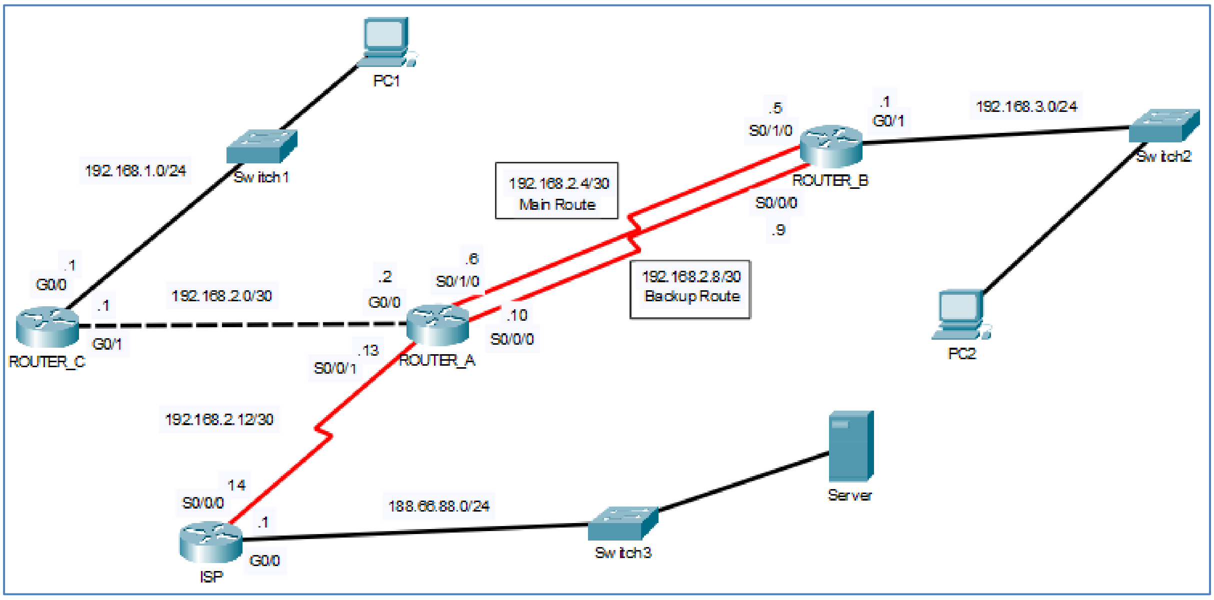
Tutorial 1

1. Pass year 202601

You are appointed as a network administrator to analyse, demonstrate, and configure different types of static routes for a network topology where Internet Protocol version 4 (IPv4) addressing is implemented on all devices, as illustrated in **Figure 1-1**.

Assume that a **fully specified default static route** has been preconfigured on **ROUTER_C**, and a **default static route** has also been preconfigured on **ROUTER_A** to forward traffic to the **ISP**. Additionally, **standard static routes** have been preconfigured on the **ISP** to route packets to **PC1** and **PC2**.



With reference to **Figure 1-1**, answer the following questions.

(i) **ROUTER_B** requires a primary default route and a backup (floating) default route to forward traffic to **ROUTER_A**. Explain how the concept of **Administrative Distance (AD)** is manipulated to achieve this redundancy, and describe the router's exact behaviour when the primary link fails. (8 marks)

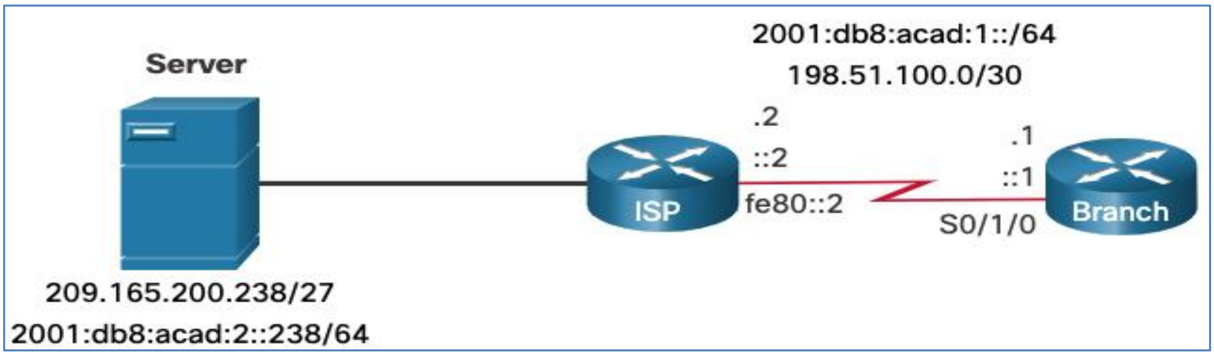
(ii) Compare the operational differences between a **standard static route** (using only a next-hop IP address) and a **fully specified static route** (using both an exit interface and a next-hop IP address) when **ROUTER_A** forwards packets to the **PC1** network. (4 marks)

(iii) Explain **TWO (2)** reasons why a **fully specified standard static route** need to be configured in **ROUTER_A**. (4 marks)

2. Illustrate in detail the **FIVE (5)** steps to obtain a Summary static route for the following networks with the same exit interface s0/0/0:

192.168.98.0, 192.168.99.0, 192.168.100.0, 192.168.101.0, 192.168.102.0

3. (i) Based on the following diagram, configure an IPv4 Static Host Route and IPv6 Static Host Route using next hop IP address in Branch to direct traffic to a Server.



(ii) Explain a host route.

(iii) Explain **TWO (2)** ways a host route can be added to the routing table.

4. a) Based on Figure 2-1, suggest the most appropriate static routing (exit interface or next hop IP address) on R11 and Edge that will allow both PC0 and PC1 to access the Internet with the minimum amount of router CPU and network bandwidth utilization. Justify your answers.

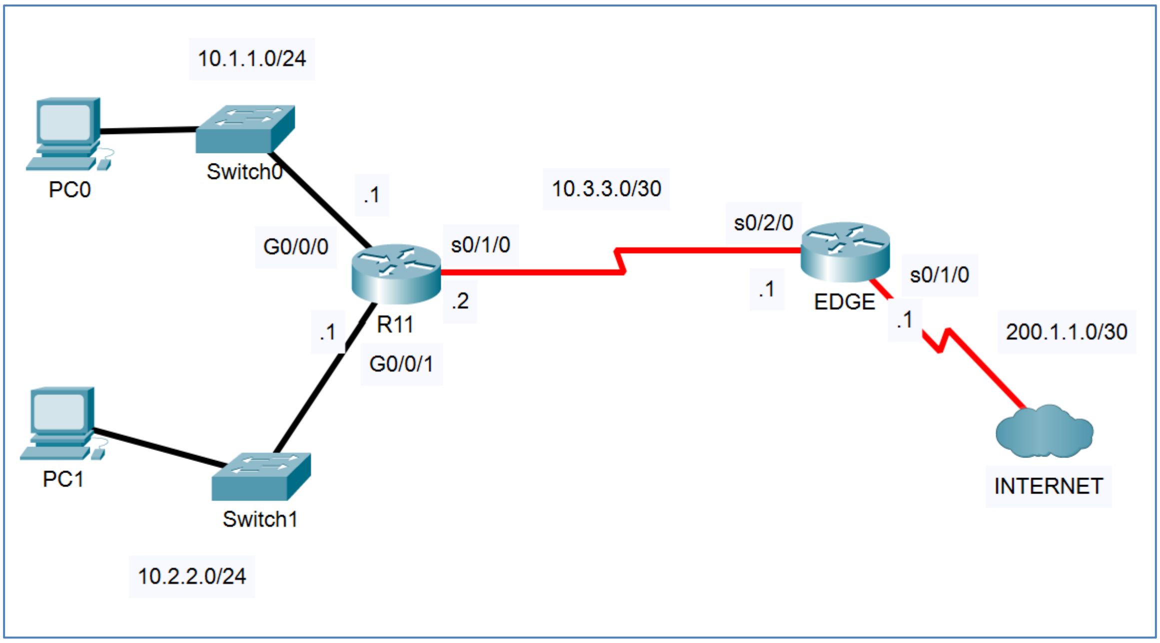


Figure 2-1 Network Topology

Q4b) Based on Figure 2-1 and Figure 2-2, EDGE will decapsulates the packet received and searches the routing table for a matching destination network entry. Explain **THREE (3)** scenarios that may occur.

```
EDGE#sh ip route
:
 Gateway of last resort is 0.0.0.0 to network 0.0.0.0

10.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
S 10.1.1.0/24 is directly connected, Serial0/2/0
C 10.3.3.0/30 is directly connected, Serial0/2/0
L 10.3.3.1/32 is directly connected, Serial0/2/0
200.1.1.0/24 is variably subnetted, 2 subnets, 2 masks
C 200.1.1.0/30 is directly connected, Serial0/1/0
L 200.1.1.1/32 is directly connected, Serial0/1/0
S* 0.0.0.0/0 is directly connected, Serial0/1/0
```

Figure 2-2 Partial output of show ip route